The two coherence conditions for a quasi-inverse are logically equivalent

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'lem:coh-equiv' (source lines 178-184); the proof environment starts on the next line after the statement, so the pairing is by adjacency.

Editorial note. No context span is retained: the statement and proof are self-contained apart from one labelled invocation, declared as an external navigation pointer.

Editorial note. The retained counter values make the counted result print with its original book number, and the chapter and section counters are set so that the retained context (where present) prints with its own original number as well.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: the equivalence of the two coherence conditions for a quasi-inverse (author label lem:coh-equiv).

Allowed standard prerequisites (editorial).

- Homotopies and quasi-inverse data. f : A -> B and g : B -> A with homotopies eta : g o f ~ id and epsilon : f o g ~ id.. They are the data of the statement.
- Naturality of a homotopy. A homotopy between composites satisfies the standard naturality square on paths.. It is used at source lines 189 and 201 to build the diagrams.
- Commutation of eta with g o f. The homotopy eta : g o f ~ id commutes with g o f in the sense of the displayed identity.. It is the labelled input used at source line 197.

Source reference. Original corollary 2.4.4: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 180–180. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Lemma 4.2.2. *For functions $f : A \to B$ and $g : B \to A$ and homotopies $\eta : g \circ f \sim \mathsf{id}_A$ and $\epsilon : f \circ g \sim \mathsf{id}_B$, the following conditions are logically equivalent:*

- $\prod_{(x:A)} f(\eta x) = \epsilon(fx)$
- $\prod_{(y:B)} g(\epsilon y) = \eta(gy)$

Proof. It suffices to show one direction; the other one is obtained by replacing A, f, and η by B, g, and ϵ respectively. Let $\tau : \prod_{(x:A)} f(\eta x) = \epsilon(fx)$. Fix $y : B$. Using naturality of ϵ and applying g, we get the following commuting diagram of paths:

$$\begin{array}{ccc} gfgfgy & \xlongequal{gfg(\epsilon y)} & gfgy \\ {\scriptstyle g(\epsilon(fgy))}\Big\| & & \Big\|{\scriptstyle g(\epsilon y)} \\ gfgy & \xlongequal[g(\epsilon y)]{} & gy \end{array}$$

Using $\tau(gy)$ on the left side of the diagram gives us

$$\begin{array}{ccc} gfgfgy & \xlongequal{gfg(\epsilon y)} & gfgy \\ {\scriptstyle gf(\eta(gy))}\Big\| & & \Big\|{\scriptstyle g(\epsilon y)} \\ gfgy & \xlongequal[g(\epsilon y)]{} & gy \end{array}$$

Using the commutativity of η with $g \circ f$ (original corollary 2.4.4), we have

$$\begin{array}{ccc} gfgfgy & \xlongequal{gfg(\epsilon y)} & gfgy \\ {\scriptstyle \eta(gfgy)}\Big\| & & \Big\|{\scriptstyle g(\epsilon y)} \\ gfgy & \xlongequal[g(\epsilon y)]{} & gy \end{array}$$

However, by naturality of η we also have

$$\begin{array}{ccc} gfgfgy & \xlongequal{gfg(\epsilon y)} & gfgy \\ {\scriptstyle \eta(gfgy)}\Big\| & & \Big\|{\scriptstyle \eta(gy)} \\ gfgy & \xlongequal[g(\epsilon y)]{} & gy \end{array}$$

Thus, canceling all but the right-hand homotopy, we have $g(\epsilon y) = \eta(gy)$ as desired. □